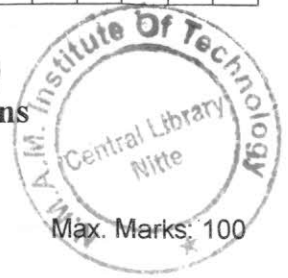


NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations
 May - June 2023

CS1001-1 – PROBLEM SOLVING THROUGH PROGRAMMING

Duration: 3 Hours



Note:

*Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.*

*Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **C was primarily developed as**
 A) Data processing language
 B) General purpose language
 C) System programming language
 D) None of the above
2. **_____ can operate on batteries and hence are very popular with travelers**
 A) Mainframes
 B) Laptops
 C) Microprocessors
 D) Hybrid
3. **C programs are converted into machine language with the help of**
 A) An Editor
 B) A compiler
 C) An operating system
 D) None of these
4. **The _____ provides pictorial representation of given problem**
 A) Algorithm
 B) Flowchart
 C) Pseudo code
 D) All of these
5. **Actual instructions in flowcharting are represented in _____**
 A) Circles
 B) Boxes
 C) Arrows
 D) Lines
6. **Which of the following is not an advantage of a flowchart?**
 A) Better communication
 B) Efficient coding
 C) Systematic testing
 D) Improper documentation
7. **The operation represented by parallelograms**
 A) Input/Output
 B) Assignment
 C) Comparison
 D) Conditions
8. **A computer that operates on digital data**
 A) remote
 B) hybrid
 C) analog
 D) digital
9. **int main()**
 {
 int a = 10.5;
 printf("%d",a);
 return 0;
 }
 A) 10.5
 B) 10
 C) 0
 D) Compilation Error
10. **Which of the following is branching statement of C language?**
 A) if statement
 B) if...else statement
 C) switch statement
 D) All of these
11. **_____ is the built in multiway decision statement in C.**
 A) for
 B) switch
 C) if
 D) while

12. What is the output of C Program.?

```
int main()
{
    int a=32;

    do
    {
        printf("%d ", a);
        a++;
    }while(a <= 30);

    return 0;
}
```

- A) 32
C) 30
B) 33
D) No Output

13. What will be the output of the following program?

```
#include<stdio.h>
int main()
{
    int a=300, b, c;
    if(a>=400)
        b=300;
    c=200;
    printf("%d%d\n", b, c);
    return 0;
}
```

- A) Garbage value, Garbage value
C) 200, 300
B) 300, 200
D) Garbage value, 200

14. If switch feature is used, then

- A) Default case must be present
C) Default case, if used, can be placed anywhere
B) Default case, if used, should be the last case
D) None of the above

15. What will be the output of the program in 16 bit platform ?

```
#include<stdio.h>
int main()
{
    int a=2, b=1, c=2;

    switch(a)
    {
        case b:
            printf("You are in b ");
            break;
        case c:
            printf("You are in c ");
            break;
        default:printf("You are in default");
    }
    return 0;
}
```

- A) You are in b
C) You are in default
B) You are in c
D) Compilation Error

16. What will be the output of the program?

```
int a=2,b=2;
if(a!=0)
    b=0;
else
    b*=10;
printf("%d",b);
```

- A) 0
C) 2
B) 20
D) None of the above

17. Which of the following function is appropriate for reading a multi-word string?
 A) printf() B) scanf()
 C) gets() D) puts()
18. What will strcmp() function do?
 A) compares the first n characters of the object B) undefined function
 C) copies the string D) compares the string
19. What is a String in C Language?
 A) String is a new Data Type in C B) String is an array of Characters with null character as the last element of array
 C) String is an array of Characters with null character as the first element of array D) String is an array of Integers with 0 as the last element of array
20. What is the Format specifier used to print a String or Character array in C Printf or Scanf function?
 A) %c B) %C
 C) %s D) %w

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Mention 6 applications of Computers and explain any 2 in detail.	8	L2	1	1
b)	Write the Algorithm and Flowchart for finding Area and Perimeter of a Circle. Area = πr^2 , Perimeter = $2\pi r$.	8	L3	1	1
2. a)	With a neat diagram, explain the Program Development Steps.	8	L2	1	1
b)	What are the Rules for forming Identifier names? Explain which of the following are Invalid identifier names and why? +point, min bal, at_a_rate, q123	8	L3	1	1
3. a)	What is the Conditional operator? Write a program to find Maximum of 2 numbers using conditional operator.	8	L3	1	1
b)	Explain the 2 methods of Data input with examples.	8	L2	1	1
Unit – II					
4. a)	What is Unconditional Branching statement? Explain the types of unconditional branching.	8	L2	2	1
b)	Write a C Program to find Sum of digits of a given number.	8	L3	2	1
5. a)	What is the difference between Entry- controlled and Exit- controlled loop? Explain the WHILE loop with example.	8	L2	2	1
b)	Write a C program to read numbers into an array and perform Linear Search. Report Success or Failure with suitable messages.	8	L3	2	1
6. a)	Explain the 4 categories of User Defined Functions in C with examples.	8	L2	2	1
b)	Write a C program using functions to read two matrices A (M x N) and B (P x Q) and to compute the product of A and B if the matrices are compatible for multiplication.	8	L3	2	1
Unit – III					
7. a)	Explain the following String functions with examples: i) strcpy ii) strcat iii) strcmp iv) strlen	8	L2	3	1
b)	Write a C program to read N integers into an array A and find the sum of elements using pointers.	8	L3	3	1
8. a)	Explain the various methods of declaring structure variables.	8	L2	3	1
b)	Write a C program to copy contents of one file to another file.	8	L3	3	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
